MODULE 19

Integration Testing and UI Test Automation

Learn how to validate your application as a whole and automate real user interactions with confidence.

TEKsystems Global Services

WORKFORCE DEVELOPMENT

MODULE 19 OVERVIEW

Validate your application as a whole and automate real user interactions with Selenium.

- Unit tests verify pieces in isolation, but real confidence comes from testing components working together and validating the UI a user actually sees. This module covers integration testing, Spring Test, and Selenium-based UI automation.

DURATION & LAB

1 Hour Theory

Integration testing, Spring Test, Selenium.

2.5 Hours Lab

Integration and UI Testing with Selenium.

Scenario

Regression-test a CRM app end-to-end.

INTEGRATION TESTING
Validate components working together

UI TEST AUTOMATION
Automate user workflows with Selenium

THE PAYOFF
Quality, reliability, faster delivery

KEY TAKEAWAY Good automation is reliable, maintainable, and provides confidence in every release.

WHY INTEGRATION TESTING MATTERS

Unit tests pass, but the application can still fail when components are wired together.

Catch Integration Bugs

- Find issues that only appear when components interact.

Validate Real Wiring

- Verify controllers, services, and repositories work together.

Test Real Protocols

- Exercise HTTP, database, and messaging boundaries.

Build Confidence

- Confirm real components behave correctly together.

Catch Config Issues

- Find misconfigured beans, profiles, and properties.

Ship Faster

- Catch issues early and release with confidence.

PRECISION: Integration tests verify that selected components, infrastructure, and application boundaries work correctly together. End-to-end tests validate complete user journeys through a running system.

KEY TAKEAWAY Integration tests catch the bugs that unit tests can't — the ones that only appear when components work together.

UNIT TESTING VS INTEGRATION TESTING

Both are essential — they test different things at different levels.

ASPECT	UNIT TESTING	INTEGRATION TESTING
Scope	A single class or method in isolation	Multiple components working together
Dependencies	Controlled to isolate behavior (mocks/stubs)	Real implementations for the boundary; unrelated externals replaced
Speed	Very fast	Slower than unit tests
Purpose	Verify business logic correctness	Verify components integrate correctly
Tools	JUnit 5, Mockito	Spring Test, @SpringBootTest, Testcontainers

PRECISION: Unit tests control dependencies to isolate behavior. Integration tests use real implementations for the selected integration boundary while replacing unrelated external systems where necessary.

"Unit tests build confidence in the code you write. Integration tests build confidence in the system you deliver."

KEY TAKEAWAY Use both: fast unit tests for logic, integration tests for confidence that the pieces fit together.

UNIT TESTING vs INTEGRATION TESTING

Both are essential and complementary in a robust testing strategy.

UNIT TESTING		INTEGRATION TESTING
Test individual units of code in isolation.	VS	Test the interaction between multiple components.
Tests a single unit of code (e.g., method, class)	SCOPE	Tests interactions between multiple units or components (e.g., services, APIs, databases)
Very fast	SPEED	Slower (involves multiple components/resources)
Usually uses mocks, stubs, or test doubles	DEPENDENCIES	Uses real dependencies (databases, APIs, message queues, etc.)
Tests internal logic, business rules, edge cases	FOCUS	Tests data flow, integration points, contract validation, end-to-end scenarios
Low (isolated, in-memory)	TEST ENVIRONMENT	High (requires multiple components and environments)
Developers (during coding)	WHO WRITES	Developers + QA Engineers (after unit tests)
Run frequently (per code change)	FREQUENCY	Run less frequently (per build, per deployment, or per release)
Finds issues in small units quickly	WHAT IT FINDS	Finds issues in component interactions, configurations, and data consistency

UNIT TESTING EXAMPLE

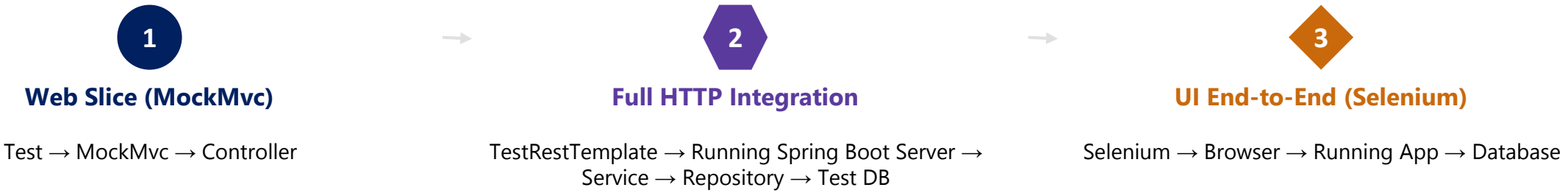
Test one unit in isolation

INTEGRATION TESTING EXAMPLE

Real interactions over network / database

INTEGRATION TESTING ARCHITECTURE

Different test boundaries use different runtime models, from a web-layer slice to a full browser end-to-end run.



KEY POINTS

- Spring-based integration tests may load a full or sliced application context depending on the boundary being tested.
- Use an in-memory database for fast basic tests only when compatibility is sufficient. Use Testcontainers with the production database engine when database-specific behavior matters.
- Use a test profile with an isolated database so integration tests don't affect production data or other test runs.
- Applications are integrated ecosystems: data flows from the UI through business logic to the database. Integration tests catch interface mismatches and data issues before they reach production.

KEY TAKEAWAY Integration tests trade some speed for confidence that real components genuinely work together.

INTEGRATION ARCHITECTURE — TESTRESTTEMPLATE

TestRestTemplate calls the running server over real HTTP, exactly like an external client would.

EXAMPLE

```
@SpringBootTest(webEnvironment =
    SpringBootTest.WebEnvironment.RANDOM_PORT)
class CustomerApiIT {
    @Autowired
    TestRestTemplate rest;

    @Test
    void createCustomer_returns201() {
        var body = new CustomerRequest("Amina");
        var resp = rest.postForEntity(
            "/api/customers", body,
            CustomerResponse.class);
        assertThat(resp.getStatusCode())
            .isEqualTo(HttpStatus.CREATED);
    }
}
```

WHY TESTRESTTEMPLATE

- Starts a real embedded server on a random port (RANDOM_PORT).
- Sends real HTTP requests — no mocked dispatcher involved.
- Exercises serialization, validation, and error mapping end to end.
- Returns a ResponseEntity so status and body are asserted together.

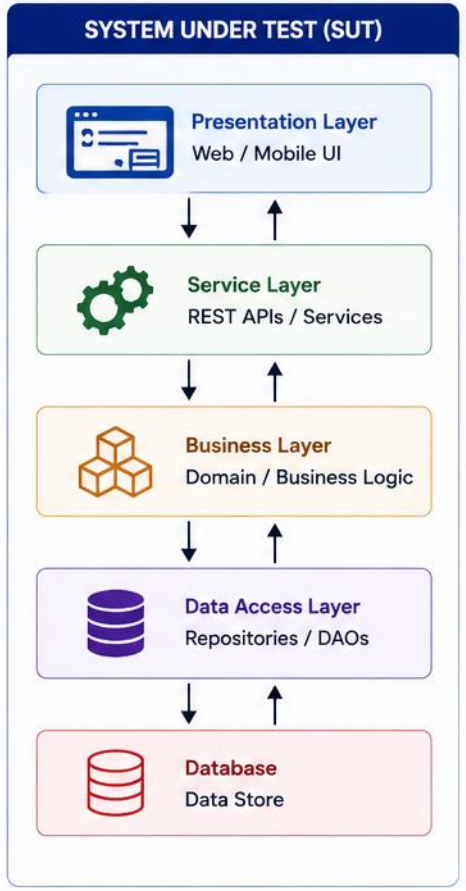
RUNS UNDER

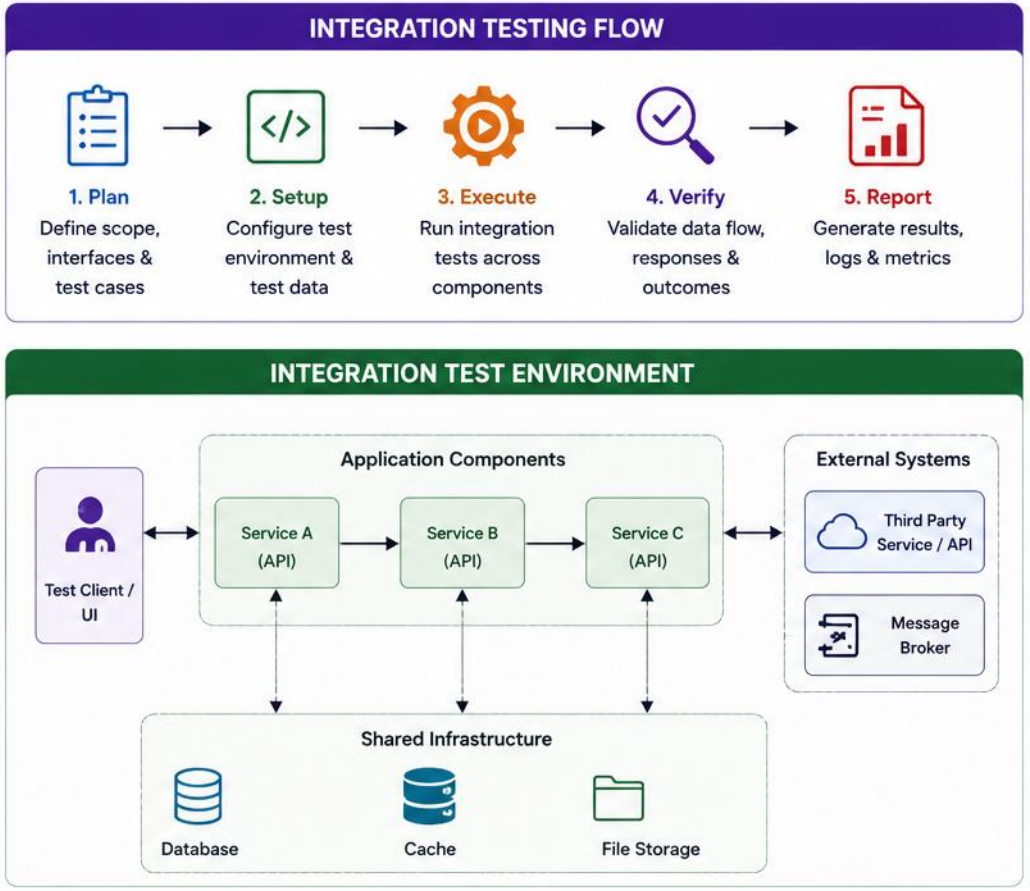
- Maven Failsafe (*IT classes), not Surefire.
- The app started with RANDOM_PORT — no external server needed.

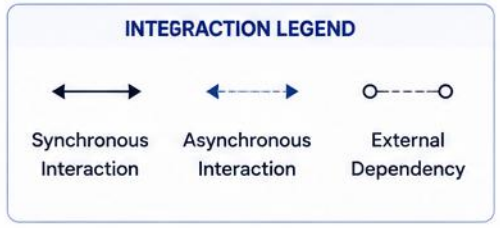
KEY TAKEAWAY TestRestTemplate turns an integration test into a real HTTP call against a running server.

INTEGRATION TESTING ARCHITECTURE

Validates interactions and data flow between integrated components in an environment close to production.





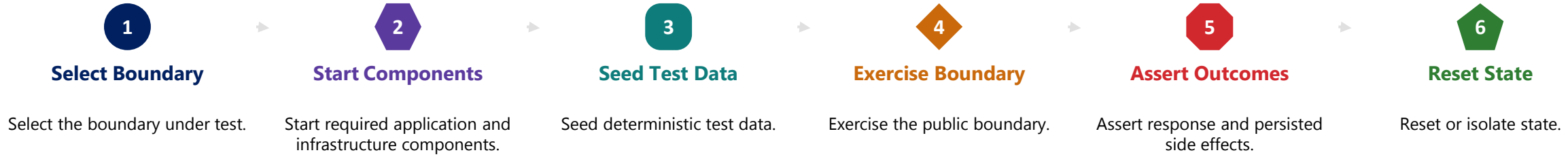


INTEGRATION TESTING ARCHITECTURE (CONT.)

A clear 3-level table for test scope, and the technical workflow for running integration tests.

TEST LEVEL	PURPOSE
`Component integration`	Multiple application components in one process
`Infrastructure integration`	Database, messaging, filesystem, HTTP adapter
`End-to-end/system`	Full running system through external interfaces

INTEGRATION TESTING FLOW (TECHNICAL)



KEY TAKEAWAY Component, infrastructure, and end-to-end/system tests each validate a different boundary — select the boundary, start the system, seed data, exercise it, assert, and reset.

INTEGRATION ARCHITECTURE — TESTCONTAINERS

@Testcontainers starts a real database in Docker so repository tests run against production-like SQL.

EXAMPLE

```
@Testcontainers
@SpringBootTest(webEnvironment =
    SpringBootTest.WebEnvironment.RANDOM_PORT)
class CustomerRepositoryIT {
    @Container
    static PostgreSQLContainer<?> db =
        new PostgreSQLContainer<>("postgres:16");

    @DynamicPropertySource
    static void props(DynamicPropertyRegistry r) {
        r.add("spring.datasource.url",
            db::getJdbcUrl);
        r.add("spring.datasource.password",
            db::getPassword);
    }
}
```

WHY TESTCONTAINERS

- Spins up a real PostgreSQL instance in Docker for the test run.
- Catches database-specific SQL and constraint issues H2 would miss.
- @DynamicPropertySource wires the container's JDBC URL into Spring.
- The container is disposable — every run starts from a clean database.

REQUIRES

- A running Docker daemon on the build machine or CI runner.
- The testcontainers-postgresql dependency on the test classpath.

KEY TAKEAWAY Testcontainers trades a little speed for confidence that queries hit a real database engine.

SPRING TEST FRAMEWORK

Three clear groups: slice tests, full application tests, and supporting configuration.

ANNOTATION	PURPOSE	TYPICAL USE
SLICE TESTS — test one layer in a partial context		
`@WebMvcTest`	Spring MVC test slice: request mapping, serialization, validation, web-layer behavior.	Testing controllers in isolation
`@DataJpaTest`	JPA/repository layer; may substitute an embedded DB unless configured otherwise.	Testing repository queries
FULL APPLICATION TESTS — the entire context, running or mocked		
`@SpringBootTest`	Loads a full or sliced context; 4 web-environment options (next slide).	Full integration tests
SUPPORTING CONFIGURATION — shape the context, not the boundary		
`@ActiveProfiles`	Activates specific Spring profiles for testing.	Selecting test vs. prod config
`@TestPropertySource`	Overrides properties for the test.	Pointing to a test database
`@AutoConfigureMockMvc`	Auto-configures MockMvc for web layer tests.	REST endpoints, no running server
`@Transactional`	Rolls back test data, but may hide commit-time behavior (see next slide).	Isolating test data between runs
`@MockBean`	Replaces a context collaborator with a test double (see next slide).	Excluding one collaborator's real behavior

SEE NEXT SLIDE: Annotation precision notes and the @SpringBootTest RANDOM_PORT example used in the Lab 19 HTTP integration tests.

KEY TAKEAWAY Choose the narrowest Spring Test annotation that gives you confidence — full context tests are powerful but slower.

SPRING TEST FRAMEWORK (CONT.)

Annotation precision notes, and the @SpringBootTest web-environment options behind the lab's HTTP tests.

@MockBean Replace selected application-context collaborators with test doubles only when the slice under test should exclude their real behavior. Do not confuse this with a pure unit test. (Spring Boot version affects the exact API — align examples accordingly.)

@WebMvcTest Loads the Spring MVC test slice for controller request mapping, serialization, validation, and web-layer behavior, while other collaborators are usually replaced.

@DataJpaTest Configure the test explicitly when the goal is to test against the production database engine, especially with Testcontainers.

@Transactional Transactional rollback helps isolate test data but may hide commit-time behavior. Use explicit commit or non-transactional tests when the production behavior depends on transaction completion.

@SpringBootTest: 4 WEB-ENVIRONMENT OPTIONS

- MOCK (default) — A mock servlet environment — no real HTTP port is opened.
- RANDOM_PORT — Starts a real server on a random free port — used by the lab's CustomerApiIT.
- DEFINED_PORT — Starts a real server on a configured, fixed port.
- NONE — No web environment or server is started at all.

RANDOM_PORT example — used in Lab 19's CustomerApiIT

```
@SpringBootTest(
    webEnvironment = SpringBootTest.WebEnvironment.RANDOM_PORT
)
```

KEY TAKEAWAY Read the fine print: @DataJpaTest's embedded database and @Transactional's rollback can hide real production behavior unless you configure them deliberately.

SPRING TEST FRAMEWORK — MOCKMVC EXAMPLE

@WebMvcTest loads only the web layer — MockMvc drives requests without a running server.

EXAMPLE

```
@WebMvcTest(CustomerController.class)
class CustomerControllerTest {
    @Autowired
    MockMvc mockMvc;

    @MockBean
    CustomerService customerService;

    @Test
    void getCustomer_ok() throws Exception {
        mockMvc.perform(get("/api/customers/1001"))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.name")
                .value("Amina"));
    }
}
```

WHY MOCKMVC

- Tests the controller layer without starting a real HTTP server.
- @MockBean replaces the service so only the web layer is exercised.
- Asserts status codes, headers, and JSON body in one fluent call.
- Much faster than a full @SpringBootTest for controller-only checks.

GOOD FOR

- Request mapping, validation, and serialization checks.
- Fast feedback loops during controller development.

KEY TAKEAWAY MockMvc gives fast, focused controller tests without paying the cost of a full running server.

SPRING TEST FRAMEWORK

Comprehensive testing support for Spring applications.

1. TEST TYPES SUPPORTED

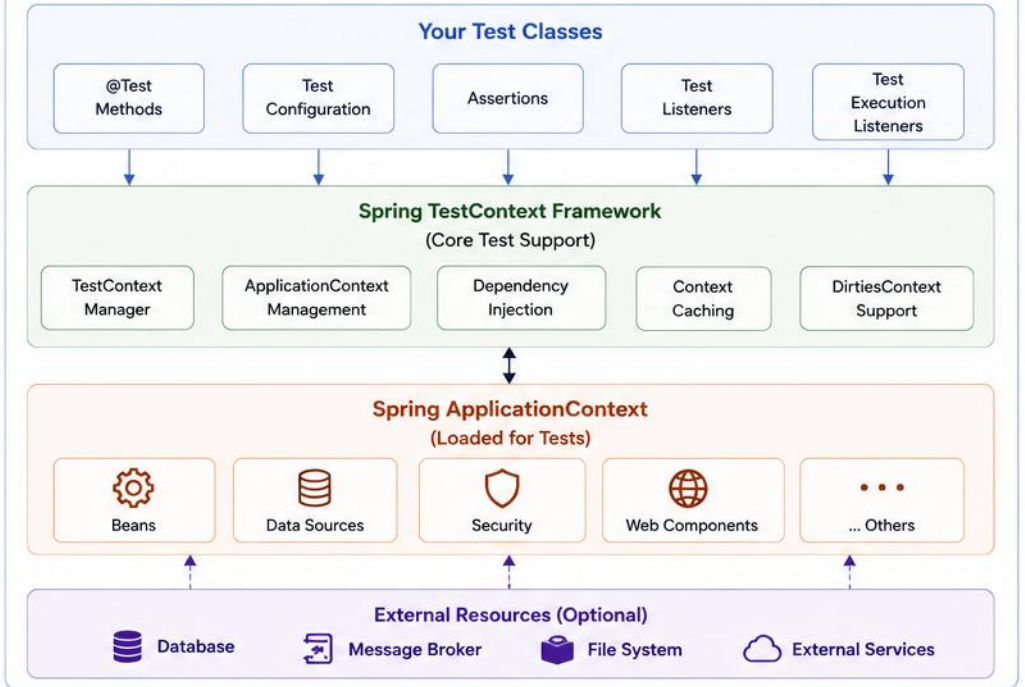
Unit Tests
Test individual components in isolation (mock dependencies)

Integration Tests
Test Spring components working together (ApplicationContext)

Web Layer Tests
Test MVC controllers, filters, views (MockMvc)

End-to-End Tests
Test full application stack with real servers / DB

2. SPRING TEST ARCHITECTURE



3. KEY ANNOTATIONS

@SpringBootTest
Load full Spring Boot context for integration tests

@WebMvcTest
Load web layer only (controllers, filters, etc.)

@DataJpaTest
Configure JPA components and test repositories

@ContextConfiguration
Specify configuration classes or locations

@TestPropertySource
Provide property sources for tests

@ActiveProfiles
Activate specific Spring profiles for testing

4. SUPPORT FEATURES

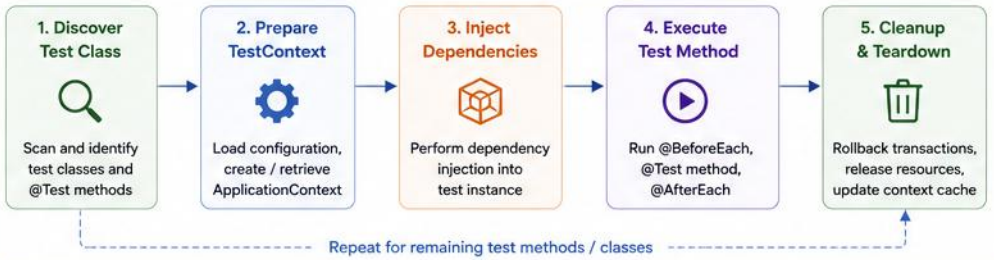
Context Caching
Reuses ApplicationContext to speed up tests

Dependency Injection
Injects beans into test classes using @Autowired, @MockBean

Transaction Management
Support for @Transactional tests with rollback

Test Execution Listeners
Hooks for setup, teardown, logging, reporting, etc.

5. TYPICAL TEST EXECUTION FLOW



6. BENEFITS

- Easy to write and maintain tests
- Leverages Spring IoC container
- Context caching for faster execution
- Rich support for web, data, security
- Extensible and highly configurable
- Improves code quality and confidence

Spring Test Framework provides a unified and flexible foundation for testing Spring-based applications at every layer.

END-TO-END TESTING CONCEPTS

End-to-end (E2E) tests validate a complete user journey through the real, running application.

E2E sits atop the test pyramid — few, slow, high-value tests above many fast unit tests and some integration tests.

Full Stack

- Exercises the entire application through external interfaces.

Real User Journeys

- Follows what a real user actually does, end to end.

Slowest, Fewest

- Few tests, but each covers a long real path.

Strong Confidence

- Provides strong confidence for critical journeys, but cannot replace lower-level tests.

Highest Cost

- Most expensive to write, run, and maintain.

Complement Other Tests

- Layer on top of unit and integration coverage, not instead of it.

PRECISION: E2E tests provide strong confidence that selected critical journeys work through the deployed stack, but they cannot replace lower-level tests.

System under test	Controlled infrastructure	Third-party systems	Production systems
Real	Real test instance/container	Sandbox / stub / contract adapter	NEVER called from tests

KEY TAKEAWAY E2E tests give strong confidence at the highest cost — use them sparingly, for critical user journeys, and keep lower-level tests too.

TRY IT YOURSELF — EXERCISE 1: TEST PYRAMID FOR NORTHSTAR CRM

Reinforces: the test pyramid you just saw — few, slow E2E/UI tests above many fast unit tests.

STEP	WHAT TO DO
Goal	Place Northstar's unit, API, and Selenium UI tests on a test pyramid before Lab 19
File	lab19-pyramid.md (in examples/module-19-exercises/notes/)
Base layer	Many fast JUnit/Mockito service-rule tests from Labs 17-18 (CustomerService validation)
Middle & top	Fewer API IT (Spring slice); few Selenium journeys — Amina ACTIVE, activate Ravi
Key concept	Fewer tests as you climb the pyramid — each layer trades speed for realism
Reference	exercises/exercise-01-test-pyramid.md

```
$ cat lab19-pyramid.md
      UI (few, slow)          -> Amina ACTIVE view, Ravi activate
   API IT (fewer)             -> CustomerApiIT (Spring slice)
Unit/Mockito (many, fast)     -> Labs 17-18 service rules
```

TRY IT NOW Complete Exercise 1 before continuing — see exercises/exercise-01-test-pyramid.md.

END-TO-END TESTING CONCEPTS

Validate the complete user journey across the entire application stack in a production-like environment.

1. WHAT IS END-TO-END TESTING?

End-to-End (E2E) testing validates **complete business workflows** from the user interface through all layers to external systems and back, just like real users.

2. E2E TEST SCOPE

User Action → UI / Frontend → Application Services → Databases → External Systems → Expected Result

Full End-to-End Coverage

3. KEY CHARACTERISTICS

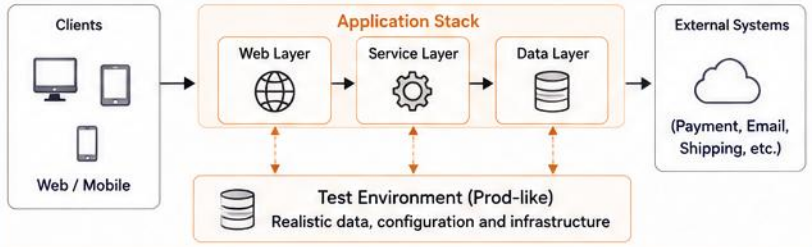
- Tests complete user workflows
- Validates integrated components
- Uses real environment & data (or close to real)
- Black-box perspective (from user view)
- Slower and more expensive than unit/integration tests

4. EXAMPLE END-TO-END FLOW (ONLINE ORDER PLACEMENT)

1. **User Browses** — Selects products and adds to cart
2. **Checkout** — User provides shipping address and payment details
3. **Order Service** — Application validates order and calculates total
4. **Data Persisted** — Order, customer and payment info saved to database
5. **Payment Gateway** — Payment request sent to external payment service and approved
6. **Confirmation** — Order confirmation email sent to the customer
7. **Order Placed** — User sees order success page with order ID

End-to-End Verification

5. E2E TEST ARCHITECTURE



6. BENEFITS

- Validates complete business workflows
- Detects issues in component integration
- Ensures data consistency across systems
- Improves user experience and reliability
- Builds confidence before production release

7. BEST PRACTICES

- Design tests around critical user journeys
- Use stable and maintainable test data
- Run in environments close to production
- Follow Page Object Model / modular design
- Execute tests as part of CI/CD pipeline
- Monitor and report results with logs/screenshots

8. WHEN TO USE E2E TESTING?

- Before major releases
- To validate critical business processes
- When multiple components work together
- To ensure production readiness
- For user acceptance validation

KEY TAKEAWAY

E2E testing ensures the application works as a whole from the end user's perspective, delivering confidence in real-world scenarios.

SELENIUM ECOSYSTEM

Selenium WebDriver is a widely adopted standard for cross-browser automation and remains common in Java enterprise testing.

Selenium IDE (Prototyping)

- Can help prototype or inspect flows, but production suites should use maintainable coded tests, stable locators, assertions, and page abstractions.

Selenium WebDriver

- Core component for writing advanced automated tests with code. This course standardizes on WebDriverManager for driver setup.

Selenium Grid

- Runs tests in parallel across multiple machines, browsers, and platforms.

HISTORICAL NOTE: Selenium RC was the earlier remote-control tool. It has been fully replaced by WebDriver.

COMMON USE CASES

- Functional and cross-browser testing of web applications.
- Regression testing of critical workflows, and data-driven, repeated test execution.
- Continuous testing in CI/CD pipelines, from UI to backend.

TOOLING CHOICE: This course standardizes on WebDriverManager, matching the lab, rather than teaching both driver-management approaches side by side. Use Selenium Manager only if your Selenium version supports the environment reliably; use WebDriverManager if the project intentionally standardizes on it.

KEY TAKEAWAY Selenium WebDriver is the modern standard: IDE helps prototype and inspect, Grid enables parallel execution, and RC is historical.

SELENIUM ECOSYSTEM — HEADLESS CHROME SETUP

WebDriverManager resolves the browser driver; ChromeOptions configures headless mode for CI.

EXAMPLE

```
WebDriverManager.chromedriver().setup();

ChromeOptions options = new ChromeOptions();
options.addArguments("--headless=new");
options.addArguments("--window-size=1920,1080");
options.addArguments("--disable-gpu");

WebDriver driver = new ChromeDriver(options);
driver.manage().timeouts()
    .implicitlyWait(Duration.ZERO);
```

WHY THIS MATTERS

- WebDriverManager downloads and matches the driver to the installed browser.
- --headless=new runs Chrome without a visible window — required for CI.
- A fixed window size keeps element layout consistent across runs.
- Configure once in @BeforeEach so every test starts from a clean browser.

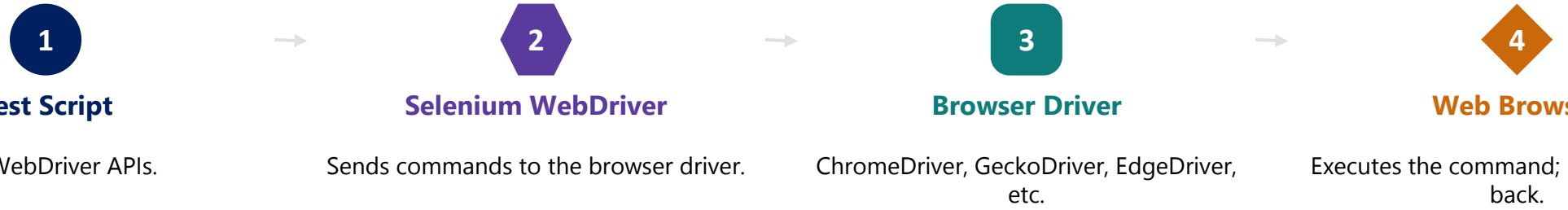
CI CHECKLIST

- Pin a WebDriverManager version compatible with the CI browser.
- Combine with an explicit window size to avoid layout differences.

KEY TAKEAWAY WebDriverManager plus headless Chrome is what makes Selenium tests reliable and fast in CI.

SELENIUM ARCHITECTURE

Selenium follows a client-server architecture: test scripts send commands to WebDriver, which controls the browser.



KEY BENEFITS

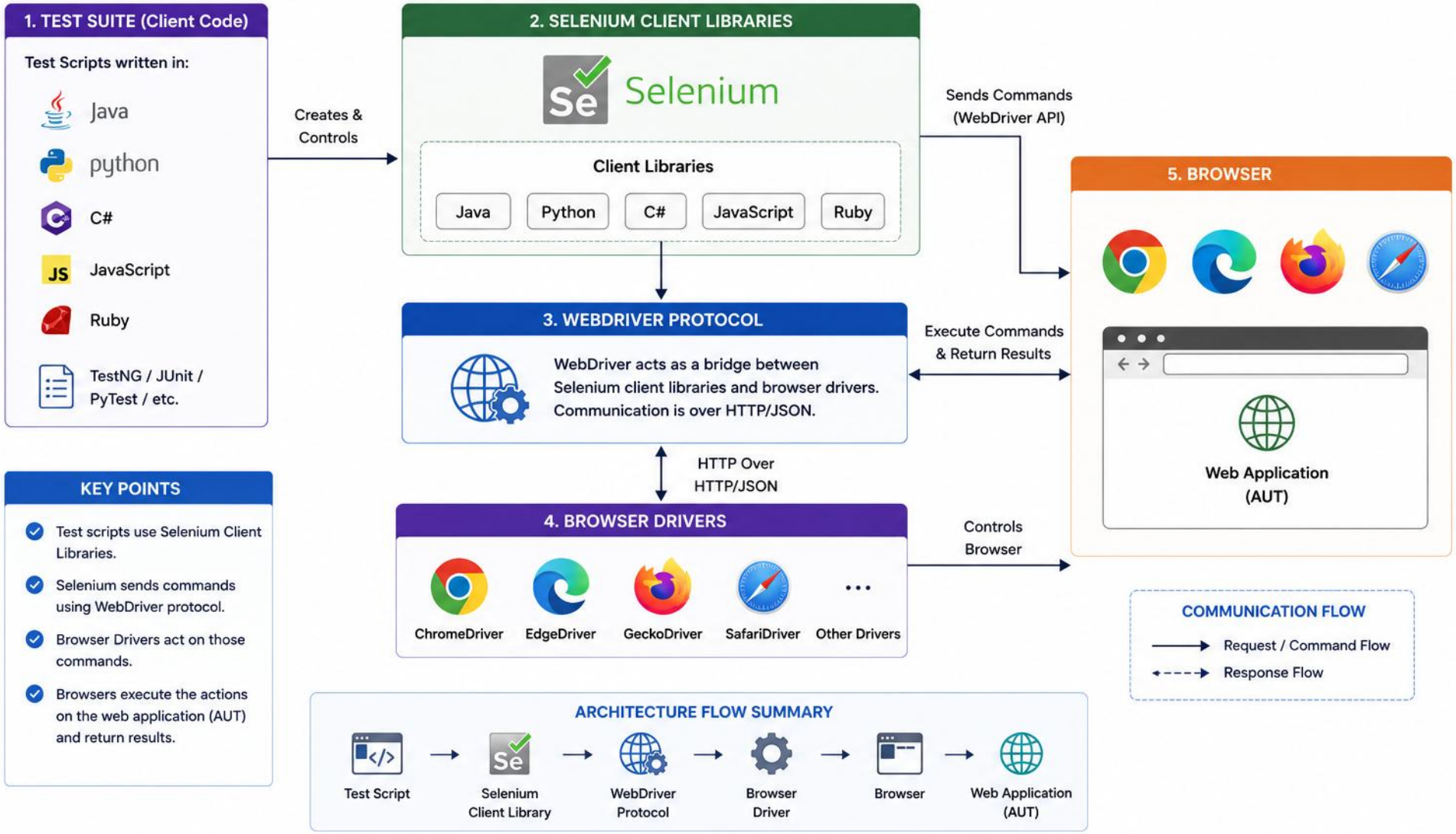
- Language independent, supports multiple browsers, enables parallel execution, and integrates with CI/CD tools.

Test code uses the Selenium WebDriver API. Selenium communicates with the browser's WebDriver implementation using the W3C WebDriver protocol.

KEY TAKEAWAY The WebDriver protocol is the bridge between your test code and the real, running browser.

SELENIUM ARCHITECTURE

Selenium automates browsers. Tests are written in a language, executed via WebDriver, which communicates with the browser to perform actions and validate results.



WEBDRIVER FUNDAMENTALS

WebDriver is the API you use to control the browser: navigate, find elements, and perform actions.

EXAMPLE

```
@BeforeEach
void startBrowser() {
    driver = new ChromeDriver(options);
}
@AfterEach
void stopBrowser() {
    if (driver != null) {
        driver.quit();
    }
}
driver.get("https://demo.ecommerce.test");
WebElement search = driver
    .findElement(By.id("search-box"));
search.sendKeys("laptop");
driver.findElement(By.id("search-button")).click();
```

CORE CAPABILITIES

- Navigate to URLs and manage browser windows.
- Find elements using locators (By.id, By.cssSelector, etc.).
- Perform actions — click, type, submit, hover.
- Wait for elements and assert on page state.

BROWSER SETUP CHECKLIST

- Headless configuration for CI runs
- Explicit window size
- Download directory, if the workflow needs one
- An explicit wait duration (not implicit waits)
- A fresh browser per test, or a carefully managed lifecycle
- No shared cookies/session unless intentional

KEY TAKEAWAY WebDriver is your programmatic remote control for the browser — every UI test is built on these fundamentals.

WEBDRIVER FUNDAMENTALS — EXPLICIT WAITS

Explicit waits are the fundamental missing piece: wait for a real condition, not a fixed sleep.

EXAMPLE

```
WebDriverWait wait =
    new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement saveButton =
    wait.until(
        ExpectedConditions.elementToBeClickable(
            By.cssSelector("[data-testid='save-customer']")
        )
    );
saveButton.click();
```

TYPES OF EXPLICIT WAIT CONDITIONS

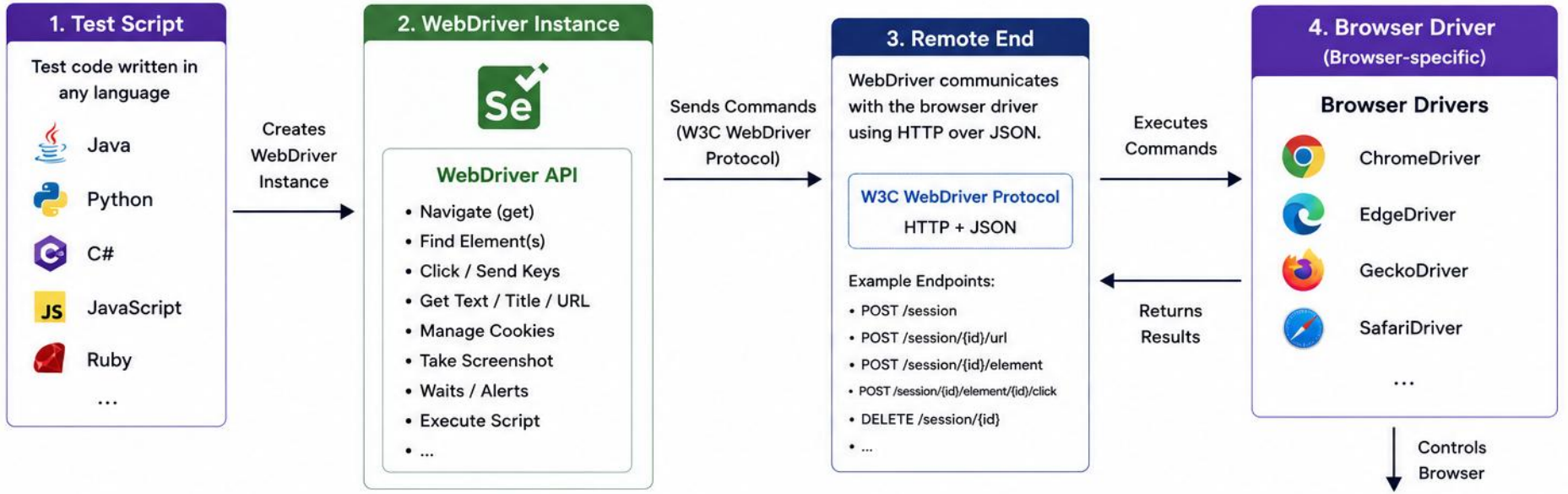
- Presence — the element exists in the DOM.
- Visibility — the element is displayed on the page.
- Clickability — the element is visible and enabled.
- URL / title change — navigation has completed.
- Custom application condition — a business state, e.g. a success alert appears.

PRECISION: Prefer explicit waits for specific state changes. Keep implicit wait at zero or a deliberately chosen minimal policy. Wait for business state — not arbitrary time.

KEY TAKEAWAY Wait for the customer row to appear, the success alert to show, or the submit button to become enabled — never for a fixed number of seconds.

WebDriver Architecture

WebDriver is the browser automation interface of Selenium. It acts as a bridge between test scripts and the browser, using the W3C WebDriver protocol over HTTP.



Key Points

- WebDriver is language neutral.
- Uses W3C WebDriver protocol over HTTP/JSON.
- Browser Drivers act as a bridge between protocol and browser.
- Enables cross-browser automation in a consistent way.
- Standardized, reliable, and widely supported.

LOCATORS AND ELEMENT SELECTION

Locators tell Selenium which element on the page to interact with.

LOCATOR	EXAMPLE	WHEN TO USE
`By.id`	By.id("login-btn")	Stable when the app provides one — not inherently "fastest"
`By.name`	By.name("username")	Good for form fields
`By.cssSelector`	By.cssSelector(".product-card")	Flexible, readable, widely preferred
`By.xpath`	By.xpath("//button[text()='Buy']")	Avoid long/absolute paths; short, intention-revealing XPath can be reliable
`By.linkText`	By.linkText("Sign Out")	Breaks under localization or copy changes — semantic locators are safer
`By.className`	By.className("form-control")	Use when a class value is stable and unique enough
`By.partialLinkText`	By.partialLinkText("Forgot")	Same fragility as linkText under copy/text changes

LOCATOR PRIORITY — BY STABILITY, NOT SPEED

1. data-testid
2. Stable ID / semantic attr
3. Accessible role/label
4. CSS selector
5. XPath (last resort)

By.className: "form-control" is valid (single class token). "form-control active" is INVALID (compound string). For multiple classes use By.cssSelector(".form-control.active").

KEY TAKEAWAY Prefer locators by stability, not speed: data-testid first, then stable IDs/semantic attributes, then CSS — keep XPath as a last resort.

TRY IT YOURSELF — EXERCISE 2: DATA-TESTID LOCATORS

Reinforces: the locator-priority list you just saw — data-testid first, XPath as a last resort.

STEP	WHAT TO DO
Goal	Propose data-testid values for the CRM elements Lab 19's Page Object will target
File	lab19-locators.md (in examples/module-19-exercises/notes/)
Element -> testid	Status badge -> customer-status; Activate button -> activate-customer; id -> customer-id
Brittle alternative	div.col-md-3 > span:nth-child(2) — flag this as brittle, never build a locator on it
Key concept	UI and tests share testids as a contract — a markup refactor shouldn't break a locator
Reference	exercises/exercise-02-data-testid-locators.md

```
$ cat lab19-locators.md
customer-status        -> status badge
activate-customer      -> activate button
customer-id            -> customer id label
// brittle: div.col-md-3 > span:nth-child(2) -- do not use
```

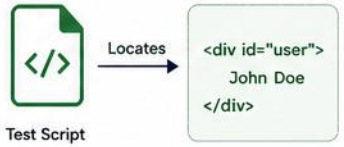
TRY IT NOW Complete Exercise 2 before continuing — see exercises/exercise-02-data-testid-locators.md.

LOCATORS AND ELEMENT SELECTION

Locators are used to identify and select web elements on a page.

1. WHAT ARE LOCATORS?

Locators are strategies used by WebDriver to find web elements in the DOM.



2. COMMON LOCATOR STRATEGIES

Locator	Description	Syntax Example	WebDriver (Java) Example	Best Used When
ID ID	Selects element by unique ID attribute.	#idValue	driver.findElement(By.id("userId"));	ID is unique and stable.
name Name	Selects element by the name attribute.	[name='nameValue']	driver.findElement(By.name("username"));	Name attribute is unique.
.class Class Name	Selects elements by class attribute.	.classValue	driver.findElement(By.className("btn-login"));	Use when class is unique.
<> Tag Name	Selects elements by HTML tag name.	tagName	driver.findElement(By.tagName("input"));	Use for generic tags (e.g., input, button).
Link Text	Selects link by exact visible text.	linkText	driver.findElement(By.linkText("Forgot Password?"));	For links with exact text.
Partial Link Text	Selects link by partial visible text.	partialLinkText	driver.findElement(By.partialLinkText("Forgot"));	For links with dynamic text.
CSS CSS Selector	Uses CSS selector syntax.	cssSelector	driver.findElement(By.cssSelector("div.container > input[type='text']"));	Powerful and flexible.
XP XPath	Uses XPath expression to find elements.	xpathExpression	driver.findElement(By.xpath("//input[@id='userId']"));	Use for complex conditions.

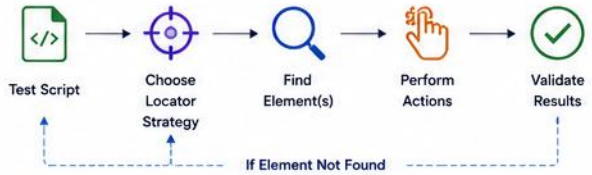
3. SAMPLE HTML SNIPPET

```
<form id="loginForm">
  <input type="text" id="userId"
    name="username" class="txt"
    placeholder="User ID" />
  <input type="password" id="pwd"
    name="password" class="txt"
    placeholder="Password" />
  <button type="submit" class="btn"
    id="loginBtn">Login</button>
  <a href="forgot.html" class="link"
    >Forgot Password?</a>
</form>
```

4. LOCATING ELEMENTS – EXAMPLES

Locator Strategy	Example (WebDriver – Java)
ID ID	driver.findElement(By.id("userId"));
name Name	driver.findElement(By.name("username"));
.class Class Name	driver.findElements(By.className("txt"));
<> Tag Name	driver.findElements(By.tagName("input"));
Link Text	driver.findElement(By.linkText("Forgot Password?"));
Partial Link Text	driver.findElement(By.partialLinkText("Forgot"));
CSS CSS Selector	driver.findElement(By.cssSelector("#loginForm .btn"));
XP XPath	driver.findElement(By.xpath("//input[@type='password']"));

5. ELEMENT SELECTION FLOW



6. BEST PRACTICES

- Prefer ID over other locators.
- Use stable and unique attributes.
- Avoid absolute XPath.
- Use relative XPath or CSS for maintainability.
- Use meaningful attribute values for automation.
- Re-validate locators when UI changes.

7. LOCATOR PRIORITY (RECOMMENDED ORDER)

High (Preferred)

1. ID
2. Name
3. CSS Selector
4. Link Text / Partial Link Text
5. Class Name
6. Tag Name
7. XPath

Low (Last Resort)

KEY TAKEAWAY

Effective element location is critical for reliable and maintainable automation scripts.
Choose the most stable and unique locator strategy for better test resilience.

AUTOMATING A MAINTAINABLE USER JOURNEY

One realistic journey, a compact workflow, and locators that hold up as the UI changes.

COMPACT WORKFLOW — JOURNEY TO MAINTAINABLE TEST

1	2	3	4	5	6	7
Understand journey	Break into steps	Identify locators	Script with WebDriver	Add validations	Handle waits	Maintain & re-run

SEE NEXT SLIDE: The Page Object Model — a CustomerFormPage example, assertion-ownership guidance, and when to stop abstracting.

KEY TAKEAWAY Automating a workflow means turning a real user journey into a repeatable, maintainable test script, validated end-to-end — not just recording clicks.

PAGE OBJECT MODEL

Separate locators and actions from scenario assertions — the pattern the lab's CustomerFormPage follows.

CustomerFormPage.java

```
public final class CustomerFormPage {
    private final WebDriver driver;
    private final WebDriverWait wait;
    private final By name =
        By.cssSelector("[data-testid='customer-name']");
    private final By email =
        By.cssSelector("[data-testid='customer-email']");
    private final By save =
        By.cssSelector("[data-testid='save-customer']");
    public CustomerFormPage enterCustomer(
            String customerName,
            String customerEmail) {
        driver.findElement(name).sendKeys(customerName);
        driver.findElement(email).sendKeys(customerEmail);
        return this;
    }
    public void save() {
        wait.until(
            ExpectedConditions.elementToBeClickable(save)
        ).click();
    }
}
```

ASSERTION OWNERSHIP: The page object exposes page state and user actions; the test owns scenario-level assertions. Small guard assertions inside page objects may confirm page readiness — avoid hiding the whole test inside page methods.

DON'T OVER-ABSTRACT: Use page objects for stable screens or meaningful components; use component objects for reusable widgets; use workflow helpers sparingly. Avoid a separate page class for every tiny fragment.

KEY TAKEAWAY A page object exposes what the page can do; the test decides what "correct" means.

TRY IT YOURSELF — EXERCISE 3: PAGE OBJECT SKETCH

Reinforces: the CustomerFormPage pattern — locators and actions live in the page, not the test.

STEP	WHAT TO DO
Goal	Sketch a CustomerStatusPage object with actions and queries, on paper, before Lab 19
File	lab19-page-object-sketch.md (in examples/module-19-exercises/notes/)
Class	CustomerStatusPage(WebDriver driver) — mirrors the lab's CustomerFormPage shape
Methods	open(customerId), readStatus(), clickActivate() — actions and queries, no asserts inside
Key concept	The page returns state (e.g. status text); the test decides what "correct" means
Reference	exercises/exercise-03-page-object.md

```
$ cat CustomerStatusPage-sketch.txt
class CustomerStatusPage(driver)
  open(customerId) / readStatus() / clickActivate()
  // status text out, activate click in -- no asserts here
// Prepare for Lab 19; don't build the full Selenium suite yet
```

TRY IT NOW Complete Exercise 3 before continuing — see exercises/exercise-03-page-object.md.

AUTOMATING USER WORKFLOWS

Automate real user scenarios by combining actions, validations and test data.

1. WHAT IS IT?

Automating user workflows means replicating real user interactions on a web application to validate business scenarios end-to-end.

2. KEY BENEFITS

- Validates end-to-end functionality
- Detects integration issues
- Improves test coverage
- Saves time and reduces manual effort
- Enables faster releases with confidence

3. COMPONENTS

- **User Actions** — Clicks, types, selects, uploads, etc.
- **Validations** — Verify expected results
- **Test Data** — Input data for different scenarios
- **Assertions** — Pass / Fail based on conditions

4. WORKFLOW AUTOMATION APPROACH

1. **Identify Workflow** — Understand and document the user scenario
2. **Break into Steps** — List all user actions and expected results
3. **Automate Steps** — Write scripts to perform actions and validations
4. **Execute & Validate** — Run the test and verify outcomes
5. **Maintain & Extend** — Update tests with application changes

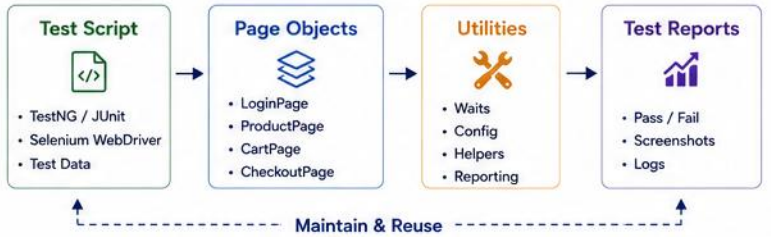
5. EXAMPLE WORKFLOW: ONLINE SHOPPING (USER JOURNEY)

1	2	3	4	5	6	7	8	9
Open Application Navigate to the website	**Search Product** Enter keyword and click search	**Select Product** Choose a product from results	**Add to Cart** Add selected product to cart	**View Cart** Open cart and verify item	**Login / Sign In** Login with valid credentials	**Checkout** Enter shipping details and continue	**Payment** Enter payment info and place order	**Order Confirmation** Verify order success message
`driver.get("https://shop.com");`	`sendKeys("laptop");` `click(searchBtn);`	`click(product1);`	`click(addToCart);`	`click(cartIcon);` `assertItem();`	`enter(username, password);` `click(loginBtn);`	`fillAddress();` `click(continueBtn);`	`enterCardDetails();` `click(payBtn);`	`assertText("Order Placed Successfully");`

6. BEST PRACTICES

- Keep workflows simple and independent
- Use Page Object Model for maintainability
- Use explicit waits for elements
- Validate each critical step
- Use meaningful data and handle exceptions
- Keep tests reusable and modular

7. SAMPLE WORKFLOW ARCHITECTURE



8. OUTCOMES

- Reliable end-to-end test coverage
- Early detection of critical issues
- Improved user experience
- Faster delivery with confidence

AUTOMATING USER WORKFLOWS = REAL USER JOURNEY + AUTOMATION + VALIDATION

Ensure your application works the way your users expect!

REGRESSION STRATEGY

A repeatable workflow, corrected regression categories, and a cadence for running each kind of test.

REGRESSION TESTING WORKFLOW

TYPE	DESCRIPTION
`Smoke Regression`	A small, fast subset confirming core functionality still works after any change.
`Change-Impact (Selective)`	Runs tests targeted at the areas a specific change could plausibly affect.
`Full Regression`	Executes the entire test suite to ensure the whole application is unaffected.
`Exploratory Regression`	Targeted manual exploration for new, changing, or highly visual behavior not yet covered by automation.

EXECUTION CADENCE

Unit tests	Component/API tests	Smoke UI tests	Full browser matrix	High-risk workflows
Every commit	Every pull request	Every PR or deployment	Scheduled / release pipeline	Release gate

KEY TAKEAWAY Regression follows identify-select-execute-compare-fix, at smoke, change-impact, full, or exploratory scope — automate the stable scenarios and explore the rest by hand.

REGRESSION STRATEGY — JUNIT 5 TAGS EXAMPLE

@Tag groups tests into smoke and regression suites that Maven can include or exclude by name.

EXAMPLE

```
@Tag("smoke")
class CustomerSmokeIT {
    @Test
    void healthEndpoint_isUp() {
        assertEquals(200, statusCode());
    }
}

@Tag("regression")
class CustomerLifecycleIT {
    @Test
    void createThenActivate_succeeds() {
        // full create -> activate -> verify
    }
}
```

WHY TAG TESTS

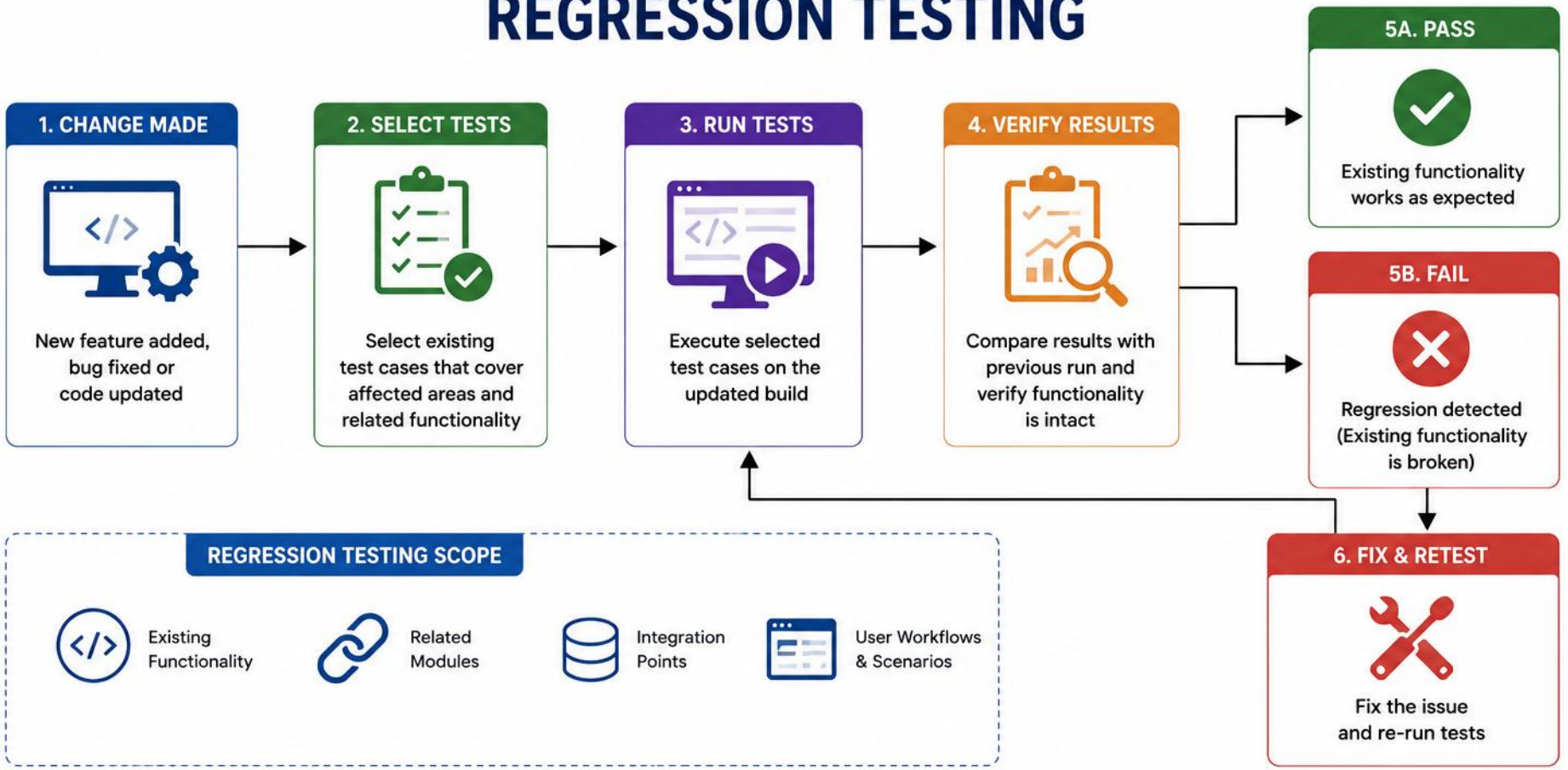
- @Tag lets Maven Surefire/Failsafe include or exclude suites by name.
- Smoke tests run on every commit; regression runs on a release gate.
- Tags map directly to the Execution Cadence table on the previous slide.
- No separate test classes or packages needed — just an annotation.

MAVEN WIRING

- mvn test -Dgroups=smoke runs only @Tag("smoke") tests.
- Surefire/Failsafe <groups> config can set this as the default.

KEY TAKEAWAY Tag tests by purpose — smoke and regression — so CI can run the right suite at the right time.

REGRESSION TESTING



BENEFITS

- Ensures existing functionality remains intact
- Detects side effects of changes early
- Reduces risk of defects in production
- Improves software quality and stability
- Supports continuous delivery and refactoring

TEST AUTOMATION BEST PRACTICES

Following these practices keeps your automation suite reliable, fast, and maintainable.

PRACTICE	WHY IT MATTERS	TOOL / TECHNIQUE
Page Object Model	Keeps locators and workflows maintainable	Page classes per screen
Explicit Waits	Avoids flaky tests from timing issues	WebDriverWait, expected conditions
Data-Driven Testing	Covers more scenarios with less code	JUnit 5 @ParameterizedTest, @CsvSource
Meaningful Assertions	Confirms outcomes, not just execution	Clear, specific assertions
Screenshots on Failure	Speeds up debugging failed tests	Automatic capture on assertion failure
CI/CD Integration	Fast feedback on every change	Surefire (*Test) + Failsafe (*IT) reports

MAVEN TEST EXECUTION (IMPORTANT): Classes named *Test run under Maven Surefire (unit tests); classes named *IT run under Maven Failsafe (integration tests) — the lab's CustomerApiIT needs Failsafe, not Surefire. Use separate Maven profiles or pipeline stages (unit, integration, ui) since Selenium tests need browser availability, display/headless config, app server lifecycle, and ports — for clearer failure diagnosis.

PARALLEL EXECUTION: Run in parallel only after tests are independent and data-isolated: unique users/IDs, separate browser instances, isolated DB state, independent ports, thread-safe page objects, no static WebDriver.

Don'ts: don't automate everything, don't use Thread.sleep(), don't ignore failed tests. Deterministic fixtures are fine — avoid shared mutable data, environment-specific values, and colliding IDs.

KEY TAKEAWAY Reliable automation avoids fixed sleeps and brittle locators — explicit waits and the Page Object Model are essential.

TEST AUTOMATION — DATA-DRIVEN TESTING EXAMPLE

@ParameterizedTest with @CsvSource covers many input cases from one test method.

EXAMPLE

```
@ParameterizedTest
@CsvSource({
    "'',      false",
    "'Amina', true",
    "'A',     true",
    "'12345', false"
})
void isValidName(String name, boolean expected) {
    assertEquals(expected,
        CustomerValidator.isValidName(name));
}
```

WHY DATA-DRIVEN TESTS

- One test method covers many input/expected-output combinations.
- @CsvSource keeps the cases readable, right next to the test.
- Adding a new case is a one-line change, not a new test method.
- Reduces duplication versus copy-pasted @Test methods per case.

OTHER SOURCES

- @ValueSource for a single primitive argument per case.
- @MethodSource when cases need to be built in code, not literals.

KEY TAKEAWAY @ParameterizedTest turns several near-duplicate tests into one table of cases.

TRY IT YOURSELF — EXERCISE 4: FLAKE AND CI NOTE

Reinforces: the explicit-waits and CI/CD guidance you just saw — flaky tests undermine trust.

STEP	WHAT TO DO
Goal	Document two flake sources and one CI constraint for the CRM's Selenium suite
File	lab19-flake-ci.md (in examples/module-19-exercises/notes/)
Flake sources	Timing/animation races; shared mutable CRM data across tests (same row, two tests)
Mitigation	Isolated Amina/Ravi fixtures, stable testids, explicit waits — never Thread.sleep()
CI constraint	Headless Chrome + WebDriverManager driver-version alignment on the build agent
Reference	exercises/exercise-04-flake-ci-note.md

```
$ cat lab19-flake-ci.md
Flake:      animation/timing races; shared mutable CRM rows
Mitigation: explicit waits, data-testid locators, isolated fixtures
CI:         headless Chrome + WebDriverManager, pinned driver version
```

TRY IT NOW Complete Exercise 4 before continuing — see exercises/exercise-04-flake-ci-note.md.

TRY IT YOURSELF — EXERCISE 5: LAB 19 PREP CHECKLIST

Reinforces: everything from the pyramid through the flake note — confirm it's ready for Lab 19.

STEP	WHAT TO DO
Goal	Confirm all five pre-lab artifacts exist; restate the pre-lab boundary before Lab 19
File	lab19-prep-checklist.md (in examples/module-19-exercises/notes/)
Artifacts to confirm	Pyramid note, page object sketch, locator contract, correlation TODOs, flake/CI note
Fixtures to recall	Amina (ACTIVE) and Ravi (PROSPECT) — the two CRM records driving Lab 19's UI stories
Key concept	Pass if all five exist; this gates readiness, it isn't the graded Lab 19 work itself
Reference	exercises/exercise-05-lab19-prep-checklist.md

```
$ cat lab19-prep-checklist.md
[ ] lab19-pyramid.md              Exercise 1
[ ] lab19-page-object-sketch.md   Exercise 2
[ ] lab19-locators.md             Exercise 3
[ ] lab19-correlation-todos.md    Exercise 4
[ ] lab19-flake-ci.md             Exercise 5
```

TRY IT NOW Complete Exercise 5 before continuing — see exercises/exercise-05-lab19-prep-checklist.md.

LAB OVERVIEW – INTEGRATION AND UI TESTING WITH SELENIUM

Extend the CRM app with Spring HTTP integration tests and a Selenium Page Object UI suite.

APPLICATION UNDER TEST

- Northstar CRM (Customer Management Platform): create and get customer records, reusing the CUS-1001 / CUS-1002 fixtures and correlation ID from Labs 17–18.
- **Database strategy: this lab keeps the existing in-memory repository with reset hooks, for time-limited training. Use PostgreSQL Testcontainers (provided setup) when the goal is realistic persistence integration.**

TECH STACK

ITEM	VALUE
`Language`	Java, Maven
`UI Automation`	Selenium WebDriver, JUnit 5
`API Testing`	Spring Test, TestRestTemplate
`Test Support`	Maven Surefire, WebDriverManager

- **LAB ARCHITECTURE:**
  API: CustomerApiIT → RANDOM_PORT → Controller → Service → Repository → Test DB
- UI: CustomerUiIT → Headless Chrome → Running App → UI/API → Repository

LAB OVERVIEW

- You will extend lab18-crm with Spring Web and Selenium: write CustomerApiIT integration tests, build a Page Object UI suite for the CRM customer form, add negative test cases, and prove the regression suite stays green across two runs.
- This is a substantial implementation project. Starter code is provided for the Spring Boot application, controller, static HTML page, Maven plugins, browser configuration, and test base class, so you can focus on API assertions, the Page Object, wait strategy, positive/negative workflows, and evidence collection.

Prerequisites: Labs 17–18 (customer domain, service rules, Mockito tests), JDK 21, Maven, Chrome/Chromium, and Git basics.

KEY TAKEAWAY This lab combines Spring HTTP integration tests and a Selenium Page Object UI suite into one regression-minded workflow for the CRM.

LAB TASKS AND DELIVERABLES

Build HTTP integration tests and a Selenium UI suite to validate the CRM app's create/get flow end-to-end.

#	TASK	DESCRIPTION	TECH
1	Environment Setup	Copy lab18-crm to lab19-crm; add Spring Web, Selenium, WebDriverManager deps.	Java, Maven, Spring Boot
2	Create/Get API	Implement create/get endpoints with X-Correlation-Id echo.	Spring Web, Spring Boot
3	HTTP Integration Tests	Write CustomerApiIT: create, get, and 404 cases via HTTP.	Spring Test, TestRestTemplate
4	Minimal UI Surface	Build customers.html with data-testid hooks for the form.	HTML, Spring Boot statics
5	WebDriver Setup	Configure WebDriverManager and headless Chrome with waits.	Selenium, WebDriverManager
6	Page Object & UI Test	Build CustomerFormPage; automate Amina's happy-path create.	Selenium WebDriver, JUnit 5
7	Negative Test Cases	Add blank-name UI validation and a missing-customer 404 case.	Selenium, Spring Test
8	Regression & Evidence	Run suite twice; capture and restore a broken-locator screenshot.	Maven Failsafe, Selenium

VALIDATION, 404s, FAILURE EVIDENCE & RELIABILITY: Browser blocks empty submission; API test (task 3) checks Bean Validation; UI (task 7) shows the mapped error. 404 is tested as both an API and a UI case. On failure, capture: screenshot, page source, console logs, URL, network logs, server logs, correlation ID, test name, browser/version. Reliability: clean-checkout pass, pass alone and as a suite, reset state, unique data, no sleep(), no order dependence.

MAVEN LIFECYCLE, CORRELATION ID & LOCATOR DEMO: CustomerServiceTest → Surefire (mvn test); CustomerApiIT/CustomerUiIT → Failsafe (mvn verify). Define Spring Boot startup, the random port reaching Selenium, browser start/stop, screenshot capture, and data reset. X-Correlation-Id: validate length/characters, generate if absent, never use for auth, block log injection. Broken-locator demo is an OPTIONAL instructor exercise, not mandatory learner work.

KEY TAKEAWAY CustomerApiIT and CustomerUiIT both pass; negative cases are covered; the regression suite stays green across two runs.

TRY IT YOURSELF — EXERCISE 6: CORRELATION TODOS (STARTER)

Reinforces: the X-Correlation-Id echo — a fill-in-the-blank starter, not built from scratch.

STEP	WHAT TO DO
Goal	Fill in the starter's correlation-header blanks for the lab's CustomerApiIT calls
File	lab19-correlation-todos.md (in examples/module-19-exercises/notes/)
Starter blanks	Header name ___; value for lab ___; must IT calls attach it? ___; UI logs it? ___
Answers to fill	X-Correlation-Id; lab-request-001; yes; yes/optional; flake idea; "no" for Actuator
Key concept	The Actuator blank must be "no" — Actuator probes are Lab 21, not this pre-lab
Reference	exercises/exercise-06-fill-correlation-header-todos.md (starter)

```
$ cat lab19-correlation-todos.md  -- STARTER, fill every blank
Header name: _____
Header value for lab: _____
IT call must attach header? _____
Flake mitigation idea: _____
Actuator in this pre-lab? _____
```

TRY IT NOW Complete Exercise 6 before continuing — see exercises/exercise-06-fill-correlation-header-todos.md (starter).

INTEGRATION AND UI TEST DEFINITION OF DONE

A checklist for a genuinely complete integration and UI test suite — not just a restatement of the objectives.

- ✓ Test boundary is explicitly defined
- ✓ Real components under test are identified
- ✓ Unrelated external systems are controlled
- ✓ Test data is deterministic and isolated
- ✓ Production data is never used
- ✓ Database technology matches the risk being tested
- ✓ HTTP tests run against a clear web environment
- ✓ Browser tests use stable test hooks
- ✓ Explicit waits replace fixed sleeps
- ✓ Page objects separate locators from scenarios
- ✓ Tests assert business outcomes, not only clicks
- ✓ Browsers and server resources are always cleaned up
- ✓ Failure evidence is captured
- ✓ Suites are separated by speed and purpose
- ✓ Tests run reliably from Maven and CI

KEY TAKEAWAY Fifteen checks, not ten objectives — that's what separates a demo from a suite you can trust in CI.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of integration testing and UI test automation.

1. A repository test passes with H2 but fails against PostgreSQL due to database-specific SQL. What should change?

A. Ignore it — H2 is close enough

B. Use an integration test against PostgreSQL (e.g. Testcontainers)

C. Remove the test since databases are unpredictable

D. Rewrite the repository to avoid SQL entirely

3. What is Selenium WebDriver's role?

A. It stores test results in a database

B. It sends commands to the browser driver

C. It replaces JUnit for assertions

D. It only tests REST APIs

2. What is the main goal of integration testing?

A. Test a single method in isolation

B. Verify components work correctly together

C. Replace unit tests entirely

D. Test only the UI

4. Which locator is generally most maintainable when the app provides a stable, dedicated test hook?

A. By.xpath with an absolute path

B. By.linkText

C. By.id, even if auto-generated and unstable

D. A selector based on a stable data-testid

KEY TAKEAWAY Integration tests verify real component wiring; Selenium WebDriver automates real browser interactions.

KNOWLEDGE CHECK (2 OF 2)

Four more questions, plus the full answer key.

5. A Selenium test uses Thread.sleep(5000) after every click. What is the better approach?

A. Increase the sleep to 10000ms to be safe

B. Wait for the specific expected UI state using an explicit wait

C. Remove all waits entirely

D. Use Thread.sleep() only in CI, not locally

7. What is a key benefit of the Page Object Model?

A. It hides bugs from testers

B. It keeps locators and workflows maintainable

C. It removes the need for assertions

D. It only works with REST Assured

6. What is the purpose of regression testing?

A. To test a feature for the first time

B. To catch regressions after code changes

C. To replace manual testing entirely

D. To test only the database layer

8. Where do end-to-end tests sit in the test pyramid?

A. At the base — many, fast tests

B. In the middle — some, moderate-speed tests

C. At the top — few, slow, high-value tests

D. They are not part of the test pyramid

ANSWER KEY

- **1-B 2-B 3-B 4-D 5-B 6-B 7-B 8-C**

KEY TAKEAWAY Reliable, maintainable automation — explicit waits, stable locators, and the Page Object Model — is what makes CI/CD safe.

"Good automation is reliable, maintainable, and provides confidence in every release."

MODULE 19 COMPLETE

Integration Testing and UI Test Automation

You can now validate applications end-to-end — from integrated components to automated real user workflows.